

LLM Cluster Naming Evaluation: MMR vs. No-MMR Entity Sampling

OWNER: Toward Unsupervised Open-World Named Entity Recognition
Llama-3.1-8B (Ollama) · BERTScore (RoBERTa-large) · Five CrossNER Domains

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Abstract

This report evaluates the effect of **Maximal Marginal Relevance (MMR)** entity sampling on LLM-based cluster naming in the OWNER system. Cluster names generated by Llama-3.1-8B-Instruct (via Ollama, $n=16$ entities) are scored with BERTScore (RoBERTa-large) against ground-truth CrossNER labels across five domains: AI, Literature, Music, Politics, and Science. Results are compared against the OWNER baseline [1].

Main finding: MMR and No-MMR produce nearly identical results for most domains (avg F_1 : MMR = 0.923, No-MMR = 0.926, $\Delta = -0.002$). No-MMR edges ahead in Music (+0.011) due to better entity-set coherence, while MMR performs slightly better in Literature (+0.007). Both conditions underperform the baseline ($F_1 = 0.944$), primarily in domains where the clustering pipeline produces less cohesive clusters in these runs.

1 Introduction

After clustering entity embeddings, OWNER assigns a human-readable name to each cluster using an LLM prompted with a small sample of entity strings. A key design choice is *how to select* those $n = 16$ example entities.

- **No-MMR (random) sampling:** entities are drawn uniformly at random from the cluster.
- **MMR sampling:** entities are selected to maximise both relevance (proximity to the cluster centroid) and diversity (dissimilarity to already-selected entities), producing a representative and non-redundant sample.

The hypothesis is that MMR yields a more informative prompt, helping the LLM infer a more accurate type label. This report tests that hypothesis using BERTScore as the semantic similarity metric between the generated name and the ground-truth CrossNER label.

2 Experimental Setup

2.1 Runs

Table 1: MLflow runs used in this report (LLM naming only).

Domain	Sampling	Run ID (prefix)	Clusters k
AI	With MMR	9e28895d	16
AI	Without MMR	22d48750	16
Literature	With MMR	89ec5ac6	14
Literature	Without MMR	a24b0559	14
Music	With MMR	d70574bc	20
Music	Without MMR	08d4cdde	20
Politics	With MMR	5b3e1bf3	22
Politics	Without MMR	56aacc85	22
Science	With MMR	eec16336	20
Science	Without MMR	14b05751	20

2.2 Ground-Truth Alignment

For each cluster, the ground-truth label is the CrossNER entity type that contributes the most entities (column-wise argmax of the confusion matrix stored as `et_test_confusion-matrix.pt`).

2.3 BERTScore

BERTScore (RoBERTa-large, `lang='en'`) computes token-level cosine similarity between the generated cluster name and the GT label. Colour coding in per-cluster tables: green $F_1 \geq 0.95$, yellow $0.90 \leq F_1 < 0.95$, red $F_1 < 0.90$.

3 Results

3.1 Domain Summary

Table 2: BERTScore P / R / F_1 for LLM naming, by domain and sampling condition. $\Delta = F_1(\text{MMR}) - F_1(\text{No-MMR})$. Baseline from [1]. **Bold** = highest F_1 per domain across new runs.

Domain	k	With MMR			Without MMR			Δ	Baseline	
		P	R	F_1	P	R	F_1		F_1	Δ_{base}
AI	16	0.919	0.925	0.922	0.919	0.925	0.922	0.000	0.929	-0.007
Literature	14	0.909	0.917	0.913	0.904	0.909	0.906	+0.007	0.954	-0.041
Music	20	0.940	0.946	0.943	0.953	0.955	0.954	-0.011	0.957	-0.003
Politics	22	0.925	0.931	0.928	0.927	0.934	0.930	-0.002	0.953	-0.023
Science	20	0.907	0.909	0.908	0.910	0.908	0.908	0.000	0.926	-0.018
Average	92	0.921	0.927	0.923	0.924	0.927	0.926	-0.002	0.944	-0.018

4 Per-Domain Per-Cluster Results

4.1 AI Domain ($k = 16$, $\Delta F_1 = 0.000$)

Table 3: AI domain — LLM naming BERTScore F_1 . Both conditions produce identical names (same cluster structure).

C#	GT label	LLM Name (both conditions)	F_1
0	algorithm	artificial intelligence/machine learning	0.883
1	researcher	computer scientist	0.855
2	organisation	organization	0.983
3	field	field of study	0.871
4	country	country	1.000
5	task	machine learning	0.893
6	metrics	metric	0.962
7	organisation	company	0.905
8	programlang	programming languages	0.882
9	university	university	1.000
10	product	machine learning library	0.874
11	conference	conference	1.000
12	product	robot	0.892
13	person	director	0.933
14	misc	award	0.853
15	metrics	metric	0.962
Mean			0.922

Note. The AI domain produces identical LLM names under both sampling conditions, indicating that the cluster entity populations are stable and MMR does not alter the representative sample meaningfully. Low scores arise from semantic gaps: *ai/ml* vs. **algorithm** ($F_1 = 0.883$), *machine learning library* vs. **product** ($F_1 = 0.874$). Three clusters reach $F_1 = 1.000$ (*country*, *university*, *conference*).

4.2 Literature Domain ($k = 14$, $\Delta F_1 = +0.007$ for MMR)

Table 4: Literature domain — LLM naming BERTScore F_1 . Shaded cells indicate the condition with higher F_1 ; ties are unmarked.

C#	GT	Name (MMR)	F_1	Name (No-MMR)	F_1	Δ
0	book	character	0.910	person	0.901	+0.009
1	misc	novels	0.856	work of fiction	0.854	+0.002
2	writer	author	0.918	author	0.918	0.000
3	organisation	organizations	0.957	organizations	0.957	0.000
4	country	country	1.000	city	0.945	+0.055
5	book	novels	0.891	novel	0.906	-0.015
6	event	film festivals	0.883	film festivals	0.883	0.000
7	writer	author	0.918	author	0.918	0.000
8	book	literary work	0.885	literary work	0.885	0.000
9	book	novels	0.891	novel	0.906	-0.015
10	misc	nationality	0.874	novelists	0.860	+0.014
11	award	award	1.000	awards	0.971	+0.029
12	location	country	0.935	country/region	0.906	+0.029
13	literarygenre	novels	0.868	novel	0.879	-0.011
Mean			0.913		0.906	+0.007

Analysis. MMR outperforms No-MMR in Literature (+0.007). The largest gain is on C4 (**country**): MMR samples more geographically diverse entities so the LLM correctly names it *country* ($F_1 = 1.000$), whereas No-MMR biases the sample toward cities and the LLM outputs *city* ($F_1 = 0.945$). Conversely, for C5/C9 (**book**), No-MMR produces *novel* (more precise) while MMR produces *novels* (plural, slight penalty).

4.3 Music Domain ($k = 20$, $\Delta F_1 = -0.011$, No-MMR better)

Analysis. Music is the domain where No-MMR most clearly outperforms MMR (+0.011). Two clusters drive this gap:

- **C1 (band):** MMR selects a diverse sample that includes album titles, leading the LLM to output *albums* ($F_1 = 0.887$). No-MMR samples more homogeneous band names, correctly yielding *band* ($F_1 = 1.000$; $\Delta = -0.113$).
- **C18 (event):** MMR’s diversity introduces non-event entities, causing the LLM to respond *multisport events* ($F_1 = 0.889$). No-MMR keeps the sample event-focused and the LLM outputs *event* ($F_1 = 1.000$; $\Delta = -0.111$).

In both cases, the cluster has a dominant entity type and MMR’s diversity *hurts* by sampling from the minority entity population.

Table 5: Music domain — LLM naming BERTScore F_1 .

C#	GT	Name (MMR)	F_1	Name (No-MMR)	F_1	Δ
0	misc	music genres	0.868	music genres	0.868	0.000
1	band	albums	0.887	band	1.000	-0.113
2	country	country	1.000	country	1.000	0.000
3	band	band	1.000	band	1.000	0.000
4	musicalartist	singer	0.905	singer	0.905	0.000
5	location	theater	0.907	theater/venue	0.901	+0.006
6	album	albums	0.944	song	0.937	+0.007
7	musicgenre	music genres	0.945	music genres	0.945	0.000
8	album	albums	0.944	albums	0.944	0.000
9	award	award	1.000	award	1.000	0.000
10	musicalartist	musician	0.930	musician	0.930	0.000
11	band	band	1.000	band	1.000	0.000
12	organisation	organization	0.983	organization	0.983	0.000
13	location	theater	0.907	theater	0.907	0.000
14	award	award	1.000	award	1.000	0.000
15	location	city	0.937	city	0.937	0.000
16	location	music genre	0.895	place	0.933	-0.038
17	organisation	organization	0.983	organizations	0.957	+0.026
18	event	multisport events	0.889	event	1.000	-0.111
19	musicalartist	musician	0.930	musician	0.930	0.000
Mean			0.943		0.954	-0.011

4.4 Politics Domain ($k = 22$, $\Delta F_1 = -0.002$)

Analysis. The overall difference is negligible ($\Delta = -0.002$). The largest single-cluster gap is C0 (location): MMR samples geographically diverse entities including ancient empires and regions, producing *geographic region* ($F_1 = 0.897$); No-MMR samples more place-name-like entities and the LLM simply outputs *region* ($F_1 = 0.951$; $\Delta = -0.054$). Five clusters (*election*, *politician* $\times 3$, *country*) reach $F_1 = 1.000$ under both conditions.

Table 6: Politics domain — LLM naming BERTScore F_1 .

C#	GT	Name (MMR)	F_1	Name (No-MMR)	F_1	Δ
0	location	geographic region	0.897	region	0.951	-0.054
1	organisation	legislative body	0.874	legislative bodies	0.873	+0.001
2	person	politician	0.904	politician	0.904	0.000
3	misc	ethnicity	0.881	nationality	0.874	+0.007
4	election	election	1.000	election	1.000	0.000
5	country	empires	0.888	historical entities	0.885	+0.003
6	organisation	organization	0.983	organization	0.983	0.000
7	politicalparty	political party	0.960	political party	0.960	0.000
8	politician	politician	1.000	politician	1.000	0.000
9	location	city	0.937	city	0.937	0.000
10	politicalparty	political parties	0.946	political party	0.960	-0.014
11	politician	politician	1.000	politician	1.000	0.000
12	misc	organizations	0.862	organizations	0.862	0.000
13	person	actress	0.875	celebrity	0.868	+0.007
14	event	film festivals	0.883	film festival	0.894	-0.011
15	organisation	university	0.891	university	0.891	0.000
16	person	actor	0.920	actor	0.920	0.000
17	event	military campaigns	0.908	military campaigns	0.908	0.000
18	politicalparty	political party	0.960	political parties	0.946	+0.014
19	event	war	0.912	war	0.912	0.000
20	country	country	1.000	country	1.000	0.000
21	location	city	0.937	city	0.937	0.000
Mean			0.928		0.930	-0.002

4.5 Science Domain ($k = 20$, $\Delta F_1 = 0.000$)

Analysis. Science is the most challenging domain ($F_1 = 0.908$) and MMR/No-MMR tie on average. The compound GT labels (`chemicalelement`, `chemicalcompound`, `astronomicalobject`, `academicjournal`) create a systematic penalty because the LLM generates natural English names (*elements*, *planet*, *journal*) that share only partial token overlap with the concatenated label strings. The largest individual discrepancy is C2 (`organisation`): MMR samples diverse institution types and the LLM outputs *professional associations* ($F_1 = 0.885$), while No-MMR samples more homogeneous entries and outputs *organization* ($F_1 = 0.983$; $\Delta = -0.098$).

Table 7: Science domain — LLM naming BERTScore F_1 .

C#	GT	Name (MMR)	F_1	Name (No-MMR)	F_1	Δ
0	misc	astronomical objects	0.849	entity names	0.861	-0.012
1	scientist	historical figures	0.841	asteroid	0.803	+0.038
2	organisation	professional assoc.	0.885	organization	0.983	-0.098
3	location	city	0.937	city	0.937	0.000
4	misc	nation	0.860	nationality	0.874	-0.014
5	astronomicalobj	planet	0.860	planet	0.860	0.000
6	chemicalelement	elements	0.884	elements	0.884	0.000
7	award	award	1.000	award	1.000	0.000
8	university	university	1.000	university	1.000	0.000
9	scientist	physicist	0.935	physicists	0.897	+0.038
10	astronomicalobj	asteroid	0.847	asteroid	0.847	0.000
11	discipline	physics	0.916	book	0.882	+0.034
12	chemicalcompound	chemical compound	0.936	chemical substance	0.924	+0.012
13	misc	proteins	0.847	biological entities	0.860	-0.013
14	person	actor/actress	0.838	actors/actresses	0.852	-0.014
15	protein	protein	1.000	protein	1.000	0.000
16	academicjournal	journal	0.892	journal	0.892	0.000
17	location	location	1.000	location	1.000	0.000
18	misc	biological processes	0.853	biological proc. (and)	0.843	+0.010
19	award	awards	0.971	awards	0.971	0.000
Mean			0.908		0.908	0.000

5 Analysis

5.1 Overall Effect of MMR on LLM Naming

Table 8: MMR effect summary per domain: $\Delta F_1 = F_1(\text{MMR}) - F_1(\text{No-MMR})$.

Domain	F_1 (MMR)	F_1 (No-MMR)	Δ	Verdict
AI	0.922	0.922	0.000	Neutral
Literature	0.913	0.906	+0.007	MMR better
Music	0.943	0.954	-0.011	No-MMR better
Politics	0.928	0.930	-0.002	Neutral
Science	0.908	0.908	0.000	Neutral
Average	0.923	0.926	-0.002	Neutral/slight No-MMR

Across 92 clusters, MMR and No-MMR produce essentially the same LLM naming quality. The global difference ($\Delta F_1 = -0.002$) is well within noise. Domain-level variation is small: Literature favours MMR (+0.007) and Music favours No-MMR (-0.011); AI, Politics, and Science are neutral.

5.2 When MMR Helps vs. Hurts

MMR improves naming when a cluster has **multiple valid entity sub-types** and diverse sampling ensures the LLM sees a representative cross-section (e.g. Literature C4: diverse geographic entities $\rightarrow$ *country*).

MMR hurts naming when a cluster has a **dominant entity type** and diversity sampling introduces minority entities that mislead the LLM (e.g. Music C1: album titles mixed into a band cluster $\rightarrow$ *albums* instead of *band*). This pattern accounts for both large negative deltas in Music (C1: -0.113, C18: -0.111).

5.3 Gap versus Baseline

Table 9: Comparison to baseline LLM naming [1]. $\Delta_{\text{base}} = F_1(\text{new}) - F_1(\text{baseline})$.

Domain	Baseline	MMR	No-MMR	Δ_{base} (best)
AI	0.929	0.922	0.922	−0.007
Literature	0.954	0.913	0.906	−0.041
Music	0.957	0.943	0.954	−0.003
Politics	0.953	0.928	0.930	−0.023
Science	0.926	0.908	0.908	−0.018
Average	0.944	0.923	0.926	−0.018

Both new conditions score below the baseline by an average of $\approx 0.018 F_1$. The gap is smallest in Music (−0.003 for No-MMR), where cluster structure appears similar to the baseline run, and largest in Literature (−0.041), where the current clustering produces more heterogeneous clusters (e.g. C1 *misc* and C13 *literarygenre* both receive low scores under all conditions).

The performance gap reflects differences in cluster cohesion between runs rather than a systematic weakness in either sampling strategy.

6 Conclusion

1. **MMR has negligible overall impact on LLM naming** ($\Delta F_1 = -0.002$ across 92 clusters). The choice between sampling strategies is secondary to cluster quality.
2. **No-MMR is marginally preferred** overall ($F_1 = 0.926$ vs. 0.923), driven by Music where high-purity clusters benefit from homogeneous (non-diverse) samples.
3. **MMR is preferred when clusters are heterogeneous**: in Literature, diverse sampling helps the LLM identify the dominant entity concept rather than latching onto a frequent but non-representative sub-type.
4. **Both conditions trail the baseline** by ≈ 0.018 , attributable to lower cluster cohesion in these experiment runs. Improving cluster purity is the primary lever for improving naming quality.

References

- [1] OWNER Research, “Cluster Naming Evaluation Report,” Internal technical report, 2025. ([cluster_naming_report.pdf](#))
- [2] P.-Y. Genest *et al.*, “OWNER — Toward Unsupervised Open-World Named Entity Recognition,” *IEEE Access*, vol. 13, pp. 50077–50096, 2025.
- [3] T. Zhang *et al.*, “BERTScore: Evaluating Text Generation with BERT,” in *Proc. ICLR*, 2020.